

Review on Impact of daylighting methods on energy efficiency in residence of different climate in India

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Abstract. Providing natural light with daylighting in the interior space of a residential building is essential to sustainable architecture. This research reviews how different daylighting strategies affect energy use in Residential buildings in various climate zones in India. Due to drastic changes in climate variations across India, this research on daylighting strategy is unique. In each region, tailored techniques are crucial for energy preservation.

In this research, papers published in Scopus, web of Science, and Google Scholar-indexed journals from 2008 to 2024 have been included. Starting from traditional daylighting techniques, which are part of India's architectural heritage, such as courtyards, windows, and reflective surfaces till advances to more contemporary techniques like light shelves, skylights, and sophisticated glazing materials that are climate-specific and maximize daylight penetration while decreasing heat gain discussed in this research.

In India's hot-dry, warm-humid, composite, temperate, and cold climates, the effectiveness of techniques varies. This study also includes shading devices, window-to-wall ratios, and building orientation. This study's findings demonstrate that daylighting strategies have universal benefits, but some must be tailored to the climate-specific context to achieve maximum energy efficiency.

This extensive analysis will be helpful for the use of energy-efficient daylighting techniques that are sensitive to the region's unique climatic challenges and cultural landscape by serving as a resource for Indian architects, designers, and policymakers.

1 Introduction

Daylighting is essential for natural light inside a building, improving energy efficiency and occupant comfort. Daylighting is critical in reducing energy use by minimizing artificial light and promoting sustainable living considering climate and culture.[1]. Vast climatic diversities influence daylighting strategies in India. In line with regional climatic conditions, optimizing natural light while ensuring energy efficiency and occupant comfort is essential for effective daylighting.[2][1].

For a country's economic growth, energy is crucial and is required for all economic sectors' survival and expansion. [3]. Efficiency in energy use is an essential consideration in residential building planning, construction, and operation. [4] The need for artificial lighting and air conditioning is minimized by using daylighting, which significantly improves energy efficiency. An appropriate design strategy can save 50%

on energy use [5] which is essential for India's urban housing and sustainable development goal[6]

The goal of this study is to evaluate the effect of daylighting methods on energy savings in residence of different climate zones in India by minimizing the need for artificial lighting through daylighting. A comparative assessment of the effects of daylighting strategies on the energy efficiency of residential buildings in India is included in the review, along with a methodical analysis of the literature and data synthesis from empirical studies. The literature screened from review from Following keywords: daylighting, residence, energy efficiency, climate zone and India. A comprehensive review of 36 papers published between 2008 and 2024 was conducted, encompassing works indexed in Scopus, Web of Science, Google Scholar, and journals from Springer and Sage Publications. Studies were chosen on the basis of focus on daylighting methods used in India, studies specific to various climates in India, research on daylighting in residential buildings, and studies on energy efficiency

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2 Daylighting methods used in residential buildings in India.

Table-1. Daylighting Methods in various climate in India

Literature	Daylighting Method	Parameter studied	Observation and Conclusion	Tool/Method	Climate
(Das & Paul, 2015)[7]	Window	Orientation	Angular buildings improve daylighting and reduce energy consumption compared to linear layouts. Proper orientation, window design, and maintenance enhance illumination, with potential energy savings from using LED lighting.	LOWLUX, MATLAB	Warm and Humid
(Bantanur & Bharathish, 2016)[8]	window	Opening, window to floor area ratio	The study looked at the daylighting of rural houses and discovered that traditional materials, small windows, and room proportions resulted in poor light distribution and uneven illumination throughout, particularly in the kitchen and bedrooms.	ECOTECH-2011	Hot and dry
(Dabe & Dongre, 2018)[9]	Window	CAWR (Carpet to window area ratio)	In residential buildings, CAWR affects window size, daylight and thermal comfort	ECOTECH 2011	Hot and dry
(Bardhan & Debnath, 2016)[10]	Window	WWR	The study proposes daylight-inclusive building byelaws using UDI and EMM to enhance energy efficiency.	DIVA v3.0 for Rhinoceros	Warm and humid
(Kumar & Kranthi, 2017)[11]	Window	Setback, opening size, obstruction distance	Window size, location, and obstructions play an important role for indoor illumination	Field measurement using Testo 480 and Testo 540	Warm and humid
(A. Gupta, 2017)[12]	Window, skylight	Building Orientation	The paper emphasises minimising environmental impact through the use of solar energy and recycled rainwater, improving ventilation, utilising locally produced materials for an energy-efficient green building, and lowering the consumption of natural resources.	Case study Literature review	Cold

(Chadalavada, K et al, 2017)[13]	Window Courtyard	Form of the building	Courtyard transformation in Andhra Pradesh vernacular houses improved thermal comfort by 30% but reduced indoor lighting by 79%. Artificial lighting or PV panels are recommended to enhance lighting levels.	IES-VE Analysis	Cold
(Dabe & Dongre, 2018)[14]	Window	Depth of offsets	This research explores how offset depth in residential buildings affects daylight levels and thermal comfort in hot, dry climates, offering design guidelines for optimizing these factors in liveable spaces.	Light Sensors (LI)-210 COR Ecotect 2011 Daysim	Warm and humid
(Nair et al., 2018)[15]	Window	Local Climate	The study recommends revisiting Kerala's climate zone classification and developing tailored design methodologies to address differences in daylight and thermal performance across the state's regions.	DIVA for Rhino and Grasshopper.	Composite
(Patki & Patki, 2018)[16]	Window	Opening design	Proper design and careful decision-making regarding daylighting parameters will help create naturally daylit space and save energy and energy costs due to artificial lighting.	Ecotect	Hot and dry
(Kranti Kumar & Kranthi, 2019)[17]	Window	Window size	Daylighting efficiency in residential buildings depends on location, window size, design, and environmental factors	Literature review	Warm and humid
(Verma & Bano, 2021)[18]	Courtyard	Methodology	Experimental models are commonly used for daylight analysis, and climate-based simulations can replace the daylight factor in overcast sky conditions.	Literature review	Warm and humid
(Chaudhary et al., 2022)[19]	Window, chimney	Chimney height optimization	The suggested chimney enhances daylighting and ventilation, which enhances the quality of the air in small slum dwellings.	CFD- IESVE version 2021, Daylight-Design Builder	Warm and humid

(Shanthi Priya & Kalaimathy, 2022)[20]	Courtyard Wind Catcher Clearstory Tiled roof	Illumination level, daylight factor, the uniformity ratio	Compared to tiled roof houses, clearstory houses, courtyard houses, and wind catcher houses achieve visual comfort. However, a clearstory house achieves glare-free light, average daylight factor, and needed threshold value.	Design Builder	Warm and humid
(Singal, 2022)[21]	Courtyard, Opening, Glass roofs	multisensory interiority	The article explores multisensory interiority in Agraharams, highlighting cultural rootedness, spatial organization, and sensory experiences.	Review	Composite
(Luthra & Sundaram, 2022)[22]	Window, Louver	External shading system	External movable louvre systems for shading windows improve interior quality while using less energy.	ladybug and honeybee plugins	Warm and humid
(Kalaimathy et al., 2023 [23])	Window	VLT in glazing. WWR	A WWR of 16% with a VLT of 0.62 and a 20% WWR with a VLT of 0.52 may satisfy daylighting requirements, per LEED v4.1 and ECBC.	Design Builder	Warm and Humid
(Chadalavada, 2024)[24]	Window, skylight, courtyards	flat roof with reflective paint coating	An 80-year-old heritage home's retrofit increased thermal comfort by 20% but decreased natural lighting by 25%, underscoring the necessity of Amaravathi's sustainable design strategy being balanced and context-specific	Revit	Warm and humid

Courtyard, window, skylight, chimney, and clearstory window are some of the prominent daylighting methods used with varying parameters.

2.1 Courtyard

Courtyards serve multiple functions in climate-responsive architecture: they act as natural light wells, increasing daylighting on small lots, and they are regarded as one of the best passive design techniques for controlling indoor air quality and temperature[25]. Courtyards with light coloured high reflective material and a larger aspect ratio help to maximise daylight. The orientation and depth of the courtyard, as well as the height of the surrounding walls all contribute to

controlling daylight[26]. Courtyards centrally located for natural light in adjacent rooms, enhancing illumination on both ground and upper floors efficiently[27]. Courtyard size varies with latitude, smaller near equator, larger towards poles for optimal daylight[28].

Courtyards optimize daylight using reflective materials, aspect ratio, orientation, depth, wall height, and size variation by latitude

2.2 Window

The demand for artificial light sources stems from variations in indoor light caused by window size, obstructions, time, room dimensions, lighting, and orientation[11]. Raising window placement on walls,

glazing with a 16% window-to-wall ratio and 0.62 visible light transmittance ensures ample sunlight and good daylighting without glare in warm and humid climate[29]. Optimal daylighting at South-East, 50% WWR increases ambient light by 63% in warm-humid areas[30]. Daylighting efficiency in residential buildings depends on location, window size, design, and environmental factors.

2.3 Skylight

Ceiling-mounted skylights are a great alternative to light pipes when it comes to adding natural light to interior spaces. Additionally, light pipes can be used in addition to this. A "natural view" of the sky is provided by skylights, which help lower the amount of electricity used for lighting and heating. The effects of artificial skylights, both visible and unseen, have been the subject of several studies[31]. Skylights provide glare-free light, reduce heat gain, and can significantly conserve energy in buildings. For sustainable design they play a crucial role by lowering emissions and enhancing indoor comfort[32].

2.4 Chimney

Integrating the chimneys into the design can improve ventilation and daylighting in single-story dwelling units.[19].

3 Impact of Daylighting Methods on energy efficiency

Building industry in India being responsible for roughly 33 percent of the country's overall use of electrical energy[10]. Indoor lighting consumes 42% of all household electrical energy in India[30]. Buildings energy consumption can be reduced by utilising daylighting to lessen their reliance on electric illumination[33].

Daylighting, or the use of natural light to illuminate building interiors, has long been a key component of energy-efficient architecture. In India's varied climate, daylighting has a significant and multifaceted impact on energy efficiency in residential buildings. The paper investigates the role of daylighting methods in increasing energy efficiency in various Indian climates, drawing on recent research.

A building's orientation and fenestration design are critical to maximising daylighting benefits. In India, where the sun's trajectory varies significantly between seasons, careful planning can ensure that daylight is used effectively all year. According to research, strategic orientation and window design can help to make buildings more energy efficient by optimising daylight use[1].

(J. Mardaljevic et al,2011) Three aspects of daylighting in residential buildings are examined in this study: non-

visual effects (NVE), energy savings from skylights, and daylight provision. The relevance of occupancy, building orientation, and climate is highlighted by key findings. The study highlights ongoing research to improve daylight metrics and introduces the sombrero plot for visualising circadian effects[34].

(I. Acosta et al,2016) To assess energy savings in residential buildings, the study examines daylight dynamic metrics in relation to window design, size, position, and room reflection. Important findings show that window size, shape, orientation, and position have a significant impact on energy savings and daylight autonomy, especially in the back of the room. Weather also affects daylight performance[35].

(K. Kalaimathy et al,2023) The study evaluated daylighting performance in warm, humid climates using UDI, ASE, and SDA metrics. Certain visible light transmittance (VLT) and window-to-wall ratios (WWR) were found to be compliant with LEED v4.1 and ECBC requirements. The recommendations include increasing sustainability and energy efficiency, as well as improving window design to balance daylight and reduce glare[29].

(I. Bournas ,2021) The study investigates Sweden's daylight regulations and concludes that daylight criteria, despite being non-mandatory, have not improved residential building performance. The current glazing-area method is frequently inapplicable, and while the two assessment methods generally agree, there are discrepancies that could allow for compliance manipulation. The study suggests making the daylight criteria mandatory[36].

4 Conclusion

This study concludes by emphasizing daylighting techniques' significant impact on energy efficiency in residential buildings across India's diverse temperature zones. Implementing appropriate daylighting strategies can significantly reduce the requirement for artificial illumination. These solutions include optimising window placement, using reflective materials, and using sophisticated glazing technologies. By allowing natural light into interior areas, courtyards increase daylighting in homes and reduce the demand for artificial lighting. Shading mechanisms and high-performance glass proved particularly effective in hot and dry climates for limiting heat gain and optimizing natural light. However, larger windows and skylights helped with passive solar heating and illumination in colder climates. Future research should focus on the long-term benefits of these strategies and their adaptability to shifting climate conditions. Overall, the adoption of personalised daylighting systems presents a viable option for reducing energy consumption and increasing environmental sustainability in India's varied climate. Future studies can explore integration of daylighting methods with smart technologies, smart windows,

automated shading to optimize daylighting as well as maintain energy efficiency and comfort in residential buildings.

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